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11:33 AM



1 site



I cannot watch video clips directly because I do not have eyes or a way to play video files.

However, I can read the text data and descriptions linked to the video to tell you what it is about.

The video you shared is titled "**Emily Riehl -- Human Mathematicians in the Age of AI**". [Link](#)

What the Video Covers



category theory library and the
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project, called the [Infinity Cosmos](#)
ists to write math code.

is usually lower quality than code

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cup interviews with Emily Riehl or
What would you like to explore

How do we train mathematicians in the next five years?

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To train mathematicians in the next five years, we must **shift focus from memorizing formulas to mastering formal logic and verification tools.**

As AI tools like Lean change the landscape of mathematics, human mathematicians will transition into "editors" and "architects" of math code rather than just calculation machines.

The goal of future training is to help students write clear, logical proofs that AI can check, while using AI to automate the tedious parts of the job.

1. Integrate Proof Assistants into the Classroom

Students need to learn how to interact with AI-driven software that checks mathematical arguments.

- **Learn Lean early:** Introduce interactive proof assistants (computer programs used to check math logic) in undergraduate or even high school classes.
- **Gamify math:** Use tools like the Natural Number Game to let students learn the rules of arithmetic like a video game.
- **Code as a second language:** Treat writing a mathematical proof the same way a software engineer treats writing code.

2. Shift Focus from Calculation to High-Level Logic

Because AI can handle standard arithmetic and formula manipulation, humans must double down on deep understanding.

- **"Why," not "What":** Spend less time on calculating complex integrals by hand and more time understanding why a mathematical structure works.
- **Synthesizing concepts:** Train students to connect different branches of mathematics, such as using geometry to solve problems in algebra.
- **Spotting subtle errors:** Teach students how to audit AI-generated code, since AI often generates math that looks correct but contains hidden flaws.

3. Embrace Collaboration Over Solo Problem-Solving

Modern mathematics is becoming a team sport driven by massive digital projects.

- **Crowdsourced math projects:** Involve students in massive collaborative math projects, similar to how [Emily Riehl's Infinity Cosmos Project](#) functions.
- **Open-source contributions:** Grade students on contributing verified theorems to global math libraries.
- **Interdisciplinary skills:** Teach mathematicians how to communicate with computer scientists and AI researchers.

4. Update the Evaluation and Testing Models

The traditional model of a student sitting alone with a pencil and paper for three hours needs an upgrade.

- **Interactive exams:** Allow students to use AI tools during tests, but grade them on their ability to guide the AI to a complete, flawless proof.
- **Code review grades:** Assess students based on how well they review, critique, and fix bad math code written by an AI.
- **Oral defense:** Require students to explain the intuition and strategy behind a proof out loud to ensure they actually understand the concepts.

Traditional vs. Future Math Training

Focus Area	Traditional Training (Past)	AI-Integrated Training (Next 5 Years)
Primary Skill	Hand calculations and memorizing proofs	Formal logic, coding, and system architecture
Tool Usage	Calculators and textbooks	Proof assistants (Lean), large language models
Error Checking	Professor grades the paper weeks later	AI provides instant, real-time feedback on logic
Project Style	Working alone on homework sets	Contributing to open-source global math libraries

Would you like to explore **how the Lean proof assistant works**, or should we look into

specific universities that are already changing their math degrees?



break it down to two categories: math educators and math researchers.

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To train mathematicians successfully over the next five years, we must split our strategy into two distinct areas.

Math educators must change *how* students learn foundational concepts, while **math researchers** must adapt to a world where AI changes *how* new knowledge is discovered and verified.

Here is how the training breaks down for both groups.

1. Math Educators: Redesigning the Classroom

The goal for educators is to move students away from memorizing steps and toward mastering logic, debugging, and high-level strategy.

- **Coding the Proofs:** Replace traditional paper homework with coding assignments. Students will write math using computer languages like **Lean** to get instant feedback on whether their logic is sound.
- **Teaching "AI Auditing":** Give students math problems solved by AI that contain hidden, subtle errors. Train them to spot, explain, and fix those mistakes.

- **Testing for Intuition:** Change exams from "solve this equation" to "explain the strategy behind this solution." Use oral tests or interactive coding exams where AI is allowed, but the human must guide it.
- **Gamifying Logic:** Introduce educational math games early in high school and college. This lets students learn the rigid rules of math logic through interactive puzzles rather than dry textbooks.

2. Math Researchers: Building Digital Architects

The goal for researchers is to learn how to manage massive digital math libraries and collaborate with AI to break new ground.

- **Managing "Pull Requests":** Train young researchers to act like software editors. They need to know how to review, clean up, and accept math code sent to global libraries by both humans and AI.
- **Formulating Big Ideas:** Teach researchers how to break down massive, complex mathematical concepts into smaller pieces that an AI assistant can help verify.
- **Cross-Field Translation:** Train researchers to speak both the language of advanced math (like geometry or algebra) and the language of computer science, bridging the gap between human ideas and machine logic.
- **Collaborative Research:** Shift training away from the "lone genius" model. Focus instead on working within massive, open-source online projects to solve the world's hardest problems together.

How Their Roles Compare

Feature	Math Educators (Teaching)	Math Researchers (Discovering)
Primary Goal	Build logical thinking and basic coding skills.	Discover new math and verify global libraries.

AI's Daily Role	A digital tutor that gives students instant feedback.	A tireless assistant that checks steps and writes code drafts.
Human Superpower	Explaining the "why" and creating fun learning paths.	Seeing the big picture and guiding the direction of a proof.
Biggest Shift	Grading the <i>thinking process</i> rather than the final answer.	Learning to <i>review and edit</i> code rather than just writing it by hand.

If you want to dive deeper, we can explore **specific tools educators are using today**, or we can look at **how a math researcher uses Lean in their daily work**. Which path would you like to take?

